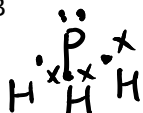


2010 9647 H2 Chemistry

Paper 1 Answer Key:

1	B	11	B	21	D	31	A
2	B	12	D	22	B	32	B
3	B	13	C	23	B	33	A
4	D	14	B	24	D	34	B
5	B	15	C	25	D	35	C
6	C	16	B	26	A	36	B
7	A	17	C	27	B	37	C
8	A	18	C	28	C	38	A
9	D	19	B	29	C	39	A
10	A	20	C	30	D	40	C

Paper 1 Worked Solutions

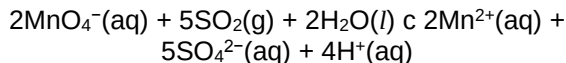
Q1 Answer: B Atomic radius decreases across a period. Since Br, K, Kr and Sc are all in the same period and K is in Group I, K would have largest atomic radius.	Q2 Answer: B Amount of $\text{CO}_2 = 1.1 \times 10^7 \div 22.4 = 491100 \text{ mol}$ Amount of $\text{Na}_2\text{CO}_3 = 491100 \div \frac{3}{2} = 327400 \text{ mol}$ Mass of Na_2CO_3 produced = $327400 \times 106.0 = \underline{3.5 \times 10^7}$
Q3 Answer: B Number of neutrons in Po = $210 - 84 = \underline{126}$ Number of nucleons in X = $210 - 4 = 206$ Number of protons in X = $84 - 2 = \underline{82}$ Number of neutrons in X = $206 - 82 = \underline{124}$	Q4 Answer: D Since no chemical rxn occurs between the gecko's toe pads and the glass surface, co-ordinate bonds (dative bonds) and covalent bonds cannot be formed. Glass is made of substances that are covalent in nature (e.g. SiO_2). Hence, there cannot be any ionic bonds formed. The attraction between the gecko's toe pads and the glass surface should be a weak one. Hence, the attraction should be van der Waals' forces.
Q5 Answer: B  PH_3 has 3 bond pairs and 1 lone pair. Hence, it has a trigonal pyramidal shape with a bond angle of 107°	Q6 Answer: C Ions in an ionic lattice should be arranged such that the each ion is surrounded by ions of the opposite charge (i.e. each o must be surrounded by •).
Q7 Answer: A NH_3: intermolecular hydrogen bonds C_2H_6, CH_4, N_2: intermolecular van der Waals' forces Since the intermolecular forces of attraction between NH_3 molecules are the strongest, NH_3 deviates the most from ideal gas behavior.	Q8 Answer: A Atomisation: endothermic because covalent bonds are broken Combustion: exothermic Formation: endothermic or exothermic Solution: endothermic or exothermic
Q9 Answer: D When a reaction is catalysed, the rate constant must increase, and the activation energy must be lowered, without affecting the energy levels of the reactants and products.	Q10 Answer: A When steam condenses, it is a phase change and hence, $\Delta G = 0$. $\Delta G = \Delta H - T\Delta S$ $0 = -44000 - (373 \times \Delta S)$ $\Delta S = -118 \text{ J K}^{-1} \text{ mol}^{-1}$ Since the unit given for ΔS in the options is J K^{-1}, we need to factor in the total amount of water (ie. $n=3 \text{ mol}$) Hence, the total entropy change $= 118 \text{ J K}^{-1} \text{ mol}^{-1} \times 3 \text{ mol} = \underline{-354 \text{ J K}^{-1}}$

Q11 Answer: B

Reducing agents are oxidised. Hence, Fe^{2+} and Br^- can be reducing agents.

Since $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}}$ is **less positive than** $E^\circ_{\text{Br}_2/\text{Br}^-}$, Fe^{2+} is more easily oxidised, and is the stronger reducing agent.

Q12 Answer: D



As the reaction proceeds, H^+ is being produced. Hence, the pH of the solution will be expected to decrease until the limiting reagent (MnO_4^-) is used up.

Q13 Answer: C

	$\text{N}_2\text{O}_4 \rightleftharpoons$	2NO_2
Initial amount / mol	x	0
Change in amount / mol	-0.5x	+x
Eqm amount / mol	0.5x	x

Total amount of gas at eqm = 1.5x

$$K_p = \frac{(P_{\text{NO}_2})^2}{P_{\text{N}_2\text{O}_4}}$$

$$= \frac{\left(\frac{x}{1.5x}\right)^2}{\frac{0.5x}{1.5x}}$$

$$= \frac{4}{3}$$

Q14 Answer: B

From the question, high charge-density cations can cause the precipitation of the solid particles.

Ionic radius of Mg^{2+} : 0.065nm

Ionic radius of Al^{3+} : 0.050nm

Ionic radius of Fe^{3+} : 0.064nm

Since Al^{3+} has the highest charge and smallest ionic radius, it has the highest charge density.

SiCl_4 is not ionic. Hence, there are no cations.

Q15 Answer: C

A is wrong: P_4 does not have the highest melting point in Period 3. Si does.

B is wrong: Sulfur can also form two acidic oxides (SO_2 and SO_3).

C is correct: Phosphorus exists as P_4 .

D is wrong: except of sodium chloride, all other soluble chlorides form acidic solutions in water.

Q16 Answer: B

Check your Periodicity lecture notes.

Option A is from NaO.

Option B is from P_4O_{10} .

Option D is from SO_3 .

Q17 Answer: C

Chlorine, bromine and iodine are acting as oxidising agents when reacting with Fe.



Chlorine can oxidise Fe to Fe^{2+} ($E^\circ_{\text{cell}} = +1.80\text{V}$). Chlorine can further oxidise Fe^{2+} to Fe^{3+} ($E^\circ_{\text{cell}} = +0.59\text{V}$). Hence, **FeCl_3** is formed.

Bromine can oxidise Fe to Fe^{2+} ($E^\circ_{\text{cell}} = +1.51\text{V}$). Bromine can further oxidise Fe^{2+} to Fe^{3+} ($E^\circ_{\text{cell}} = +0.30\text{V}$). Hence, **FeBr_3** is formed.

Iodine can oxidise Fe to Fe^{2+} ($E^\circ_{\text{cell}} = +0.98\text{V}$). However, iodine cannot further oxidise Fe^{2+} to Fe^{3+} ($E^\circ_{\text{cell}} = -0.23\text{V}$). Hence, **FeI_2** is formed.

Q18 Answer: C

Check your Transition Metals lecture notes.

Options A, B and D are properties of transition metals, not their **definition**.

Q19 Answer: B

The C–H bond in propyne is shorter than the C–H bond in ethane.

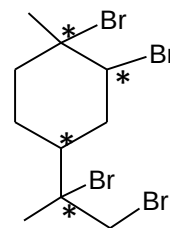
C2 in propyne is sp hybridised while C3 is sp^3 hybridised. Hence, there is $sp-sp^3$ overlap. In ethane, both carbon atoms are sp^3 hybridised.

sp hybridised C: 50% s character, 50% p character
 sp^2 hybridised C: 33% s character, 67% p character
 sp^3 hybridised C: 25% s character, 75% p character

The **higher % of s character, the shorter the bond.**
(*s orbital is spherical and hence has a shorter span as compared to the dumb bell-shaped p orbital*) Thus, **$sp-sp^3$ overlap leads to a shorter bond** in propyne.

Q20 Answer: C

When limonene is reacted with an excess of Br_2 in an inert solvent, in the absence of UV light or heat, only electrophilic addition occurs. Hence, the product formed is:



There are 4 chiral centres, as marked out by *.

Q21 Answer: D

A is wrong: Substituting a hydrogen atom on the C=C bond with a chlorine atom cannot be carried out in one step. Free radical substitution cannot bring about this change readily.

B is wrong: Fluoroalkanes do not undergo nucleophilic substitution readily as the C–F bond is very strong.

C is wrong: The $-CH_3$ group on the benzene ring is a 2,4-directing group. Hence, substitution of $-NO_2$ on position 3 of the benzene ring does not occur readily.

D is correct: Can be done using Br_2 with $FeBr_3$ catalyst

Q22 Answer: B

A is wrong: Hydrogen iodide in the aqueous form exists as H^+ and I^- ions as it is a strong acid. Thus, there are no more HI molecules which can act as electrophiles.

B is correct: This is an elimination reaction. Elimination occurs when an halogenoalkane reacts with a base. $CH_3CH_2O^-$ is a base.

C is wrong: $AlCl_3$ catalyst has to be in the solid state. In the aqueous state, it reacts with water: $[Al(H_2O)_6]^{3+} \rightleftharpoons [Al(OH)(H_2O)_5]^{2+} + H^+$ (refer to Periodicity lectures notes)

D is wrong: When chlorine is in the aqueous state, it reacts with water to form Cl^- and ClO^- ions. There are no more Cl_2 molecules to form chlorine radicals in initiation step. Hence, FRS cannot occur.

Q23 Answer: B

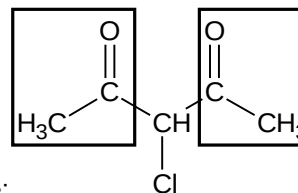
A is wrong: **A** shows an acyl chloride which will be hydrolysed with aqueous $AgNO_3$ at room temperature. This will lead to the formation of white silver chloride precipitate immediately.

B is correct: **B** shows a halogenoalkane which can only be hydrolysed to give the chloride ion only upon warming with $NaOH$.

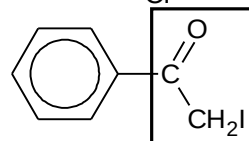
C is wrong: **C** shows a halogenoarene which cannot be hydrolysed at all.

D is wrong: **D** also shows an acyl chloride which will be hydrolysed with aqueous $AgNO_3$ at room temperature. This will lead to the formation of white silver chloride precipitate immediately.

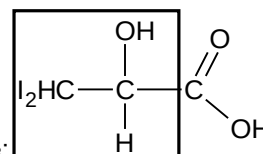
Q25 Answer: D



A can give CHI_3 :



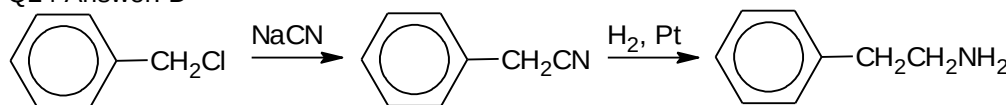
B can give CHI_3 :

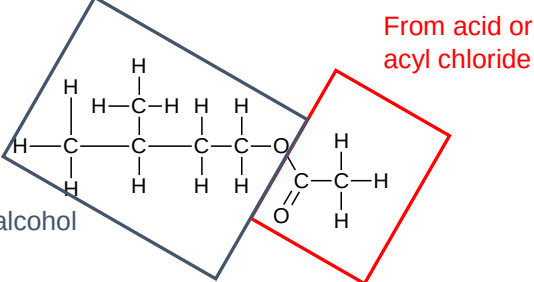


C can give CHI_3 :

D is an ester. Hence, it does not give a positive observation in the iodoform test.

Q24 Answer: D



Q26 Answer: A For X: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CHCHO} \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}(\text{HOH})$ For Y: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCOCH}_3 \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}(\text{HOH})\text{CH}_3$ For Z: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}=\text{CHCHO} \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CHCH}=\text{CHCH}(\text{HOH})$ Average hydrogen atoms incorporated per molecule = $(2 \times 2 + 2 \times 2 + 2) \div 5 = \underline{2.0}$	Q27 Answer: B Only the phenol group and the carboxylic acid groups are acidic and will be deprotonated. The alcohol group is not acidic enough (neutral) to react with sodium hydroxide.
Q28 Answer: C 3-methylbutyl ethanoate: 	Q29 Answer: C Hydrolysis with dilute hydrochloric acid will produce a carboxylic acid. Therefore, A and B are wrong. The number of carbon atoms should not change regardless of whether the nitrile is hydrolysed or reduced.
Q30 Answer: D You'd better know this...	
Q31 Answer: A For method 1, 1. Work out amount of HCl reacted with the carbonates using back titration 2. Given that the mass of the mixture is x g, work out the amounts of each carbonate in terms of x. 3. Write out the equations for each reaction 4. Use the mole ratio to work out the amount of each carbonate. For method 2, 1. Work out the total amount of CO_2 liberated 2. Follow steps 2 to 4 from method 1 For method 3, When H_2SO_4 is added, all the Ba^{2+} ions will be precipitated out as BaSO_4. From the weight of the precipitate collected, the amount of BaCO_3 can be worked out.	Q32 Answer: B 1 is correct: nitrogen has a lone pair that can be used to form dative bonds with the Mg^{2+} cation 2 is correct: Each nitrogen atom should form three single bonds. The nitrogen atoms circled forms two single bonds with the carbon atoms of the ring, and a third single bond with the Mg^{2+} cation. 3 is wrong: Although the magnesium atom is bonded to four other atoms, two of the bonds are dative from nitrogen atoms. An sp^3 hybridised atom has to form four bonds on its own. Hence, magnesium atom cannot be sp^3 hybridised.
Q33 Answer: A 1 is correct: Dissolution leads to an increased dispersion of the solute in its solution. Hence, entropy increases. 2 is correct: When solvent evaporates, the particles can now move at very high speed with great disorder and the large increase in volume results in a large increase in entropy. 3 is correct: Osmosis leads to the mixing of the pure solvent with the solution. Hence, entropy increases.	Q34 Answer: B 1 is correct: According to the first rate equation, when $[\text{HBr}]$ increases, rate decreases. 2 is correct: The rate equation for the second reaction corresponds to the overall equation. Hence, it must be an elementary reaction. 3 is wrong: Free radicals do not appear in the rate equation.
Q35 Answer: C 1 is wrong: Low solubility will make it difficult for the sulfate to be absorbed or dissolved, so it is difficult to incorporate it into human metabolism. 2 is correct: Long half lives are detrimental as it means that the harmful substance is in existence for a longer time. 3 is correct: Strontium is immediately below calcium in Group II. Hence, strontium can be absorbed into our bodies and reacted in the same way as calcium.	Q36 Answer: B Only dissolution and precipitation occurred. There was no redox reaction.

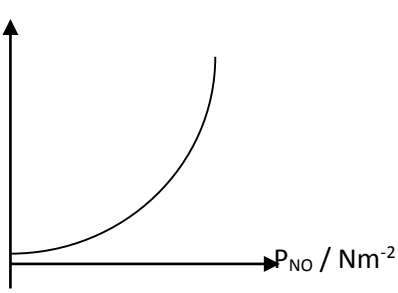
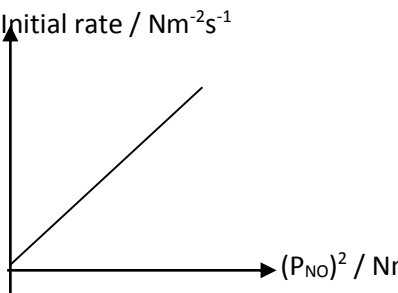
Q37 Answer: C 1 is wrong: Electrophilic substitution	Q38 Answer: A 1 is correct: A bromide ion is formed together with the carbocation 2 is correct: A bromide ion is formed at the end of the nucleophilic substitution reaction 3 is correct: A bromide ion is formed at the end of the nucleophilic substitution reaction
Q39 Answer: A 1 is correct: There is a carbonyl group 2 is correct: There is a phenol group 3 is correct: There is a phenol group	Q40 Answer: C $\begin{array}{c} \text{OH} \\ \\ \text{---C---CH}_3 \\ \\ \text{H} \end{array}$ and $\begin{array}{c} \text{O} \\ \\ \text{---C---CH}_3 \end{array}$ 1 is wrong: It does not have 2 is correct: The amide will be hydrolysed 3 is correct: The primary and secondary alcohols will be oxidised

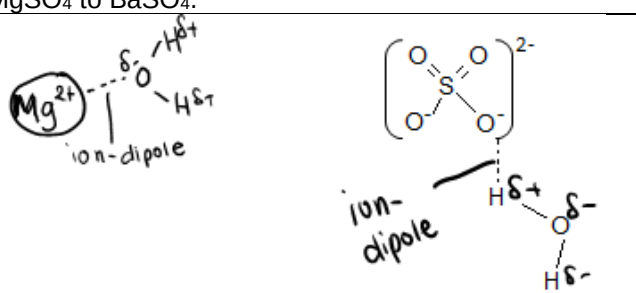
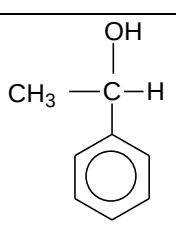
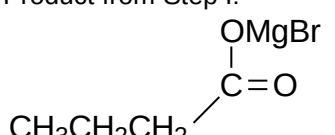
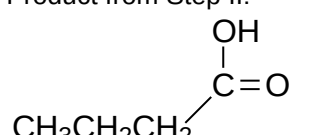
Paper 2 Worked Solutions (inclusion of SPA Planning Question)

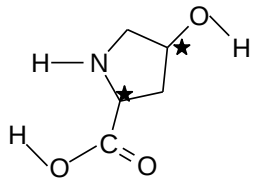
1(a)	$\text{FeO}_4^{2-} + 8\text{H}^+ + 3\text{e}^- \rightarrow \text{Fe}^{3+} + 4\text{H}_2\text{O}$ $2\text{FeO}_4^{2-} + 16\text{H}^+ + 3\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{Fe}^{3+} + 8\text{H}_2\text{O} + 6\text{CO}_2$
1(b)	The concentration of $0.100 \text{ mol dm}^{-3}$ solution of $\text{Na}_2\text{C}_2\text{O}_4$ is too diluted hence, during titration the volume of $\text{Na}_2\text{C}_2\text{O}_4$ required will exceed the 50.00 cm^3 limitation bound by the capacity of the burette. (<u>Or</u>: less than 10 cm^3 of titrant, FeO_4^{2-} which is considered inaccurate.) <i>(Tip: Students are expected to know volume measurement $< 7 \text{ cm}^3$ leads to problem of accuracy of measurement while volume $> 50 \text{ cm}^3$ would necessitate refilling of burette.)</i>
2(c)(i)	Calculation: Since 10 drops of FA1, when suitably acidified, required 32 drops of this $\text{Na}_2\text{C}_2\text{O}_4$, concentration of the $\text{Na}_2\text{C}_2\text{O}_4$ should be 3 times more concentrated, therefore 0.3 mol dm^{-3} of FA 2 is to be prepared. Mass of $\text{Na}_2\text{C}_2\text{O}_4$ required $= (0.3 \times 250/1000) \times (23.0 \times 2 + 12.0 \times 2 + 16.0 \times 4)$ $= 10.05 \text{ g} \approx \underline{10\text{g}}$ Diagram: Procedure: 1. Weigh accurately about <u>10 g</u> of solid anhydrous <u>$\text{Na}_2\text{C}_2\text{O}_4$</u> in a clean, dry <u>empty weighing bottle</u> 2. Transfer the weighed solid into a small beaker and dissolve it using <u>distilled water</u>. 3. Wash the weighing bottle with distilled water and transfer washings to the small beaker. (Step 1-3 is To ensure complete transfer of solid/accurate mass of solid used Alt: After step 2, reweigh and record the mass of emptied weighing bottle and skip step 3) 4. Transfer the solution and washings from the beaker into a <u>250 cm^3 volumetric flask</u>. 5. <u>Make up to the mark</u> with distilled water and <u>shake well</u> to obtain a homogeneous solution, label this as FA 2. (To dissolve and obtain solution)

	- Fill the burette with FA2 solution. - Pipette 25.0 cm³ of FA 1 into a conical flask and add <u>excess/10cm³ of dilute sulfuric acid</u> (To acidify with excess H₂SO₄, cannot accept HCl and HNO₃) - Titrate FA1 with FA 2 from the burette, adding <u>dropwise towards end point</u>, until the colour changes from <u>dark red to colourless / pale yellow</u>. - Repeat the titration for consistent results. (± 0.10 cm³).
1(d)	Since 2FeO₄²⁻ ≡ 3C₂O₄²⁻, by substituting the values into the expression below, the concentration of FA1 can be found: $[\text{FA 1}] = \frac{\frac{2}{3} \times \left(M \times \frac{y}{1000} \right)}{\frac{x}{1000}}$
1(e)	Wear protective gloves to handle corrosive <u>H₂SO₄ and potassium ferrate</u> or hazardous/ toxic chemicals, C₂O₄²⁻ and ferrate(VI) ions.

2(a)	Stage I: $\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr}$ Stage II: $\text{HBr} + \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{H}_2\text{O}$
2(b)	Amount of NaBr available = $35/103$ = 0.3398 mol Amount of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ available = $(25 \times 0.81)/74$ = 0.2736 mol From (a), 1 mol NaBr = 1 mol $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ Since amount of NaBr available is more than required , NaBr is in excess.
2(c)(i)	When conc H_2SO_4 is added to the aqueous reaction mixture, the dilution of the acid results in the production of heat.
2(c)(ii)	Inorganic by-product : SO_2 / Br_2 Equation: $\text{H}_2\text{SO}_4 + 2\text{HBr} \rightarrow \text{SO}_2 + \text{Br}_2 + \text{H}_2\text{O}$ Organic by-product: but-1-ene Equation: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 + \text{H}_2\text{O}$
2(d)	For the reaction to occur, the covalent bonds in the organic molecules must first be broken down , and that requires energy
2(e)	Compounds are water and butan-1-ol . Reason: they both have similar boiling points to 1-bromobutane .
2(f)	The 1-bromobutane will be on the lower layer . 1-bromobutane has a higher density than water, hence, will be at the lower layer.
2(g)(i)	Butan-1-ol
2(g)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{HCl} \rightarrow (\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}_2)^+\text{Cl}^-$
2(g)(iii)	The product is an ionic compound , therefore is able to form ion-dipole interactions for dissolution.
2(h)	Impurity: Conc HCl Equation: $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
2(i)	Water (<i>students are expected to recognise anhydrous CaCl_2 as a drying agent</i>)
2(j)	$(98-105)^\circ\text{C}$

3(a)	Ozone depletion
3(b)	 Graph 1  Graph 2 (student may plot either Graph 1 or 2, and explain accordingly in (c))
3(c)	Graph 1 When partial pressure of NO is doubled, the initial rate increased by fourfold, hence, the order of reaction with respect to NO is 2. Graph 2 Graph of initial rate against P_{NO}^2 is an upward sloping straight line passing through origin, hence, rate $\propto P_{NO}^2$ order of reaction with respect to NO is 2.
3(d)	When partial pressure of O₂ is halved the rate is halved . Thus, order of reaction with respect to O ₂ is 1 .
3(e)(i)	Rate = $k (P_{NO})^2 (P_{O_2})$
3(e)(ii)	Units: $N^{-2}m^4s^{-1}$

4(a)(i)	The magnitude of the lattice energy of BaSO ₄ will become smaller / decrease
4(a)(ii)	$ \Delta H_{latt} \propto \frac{q^+ q^-}{r^+ + r^-}$ As cationic radius increases from Mg²⁺ to Ba²⁺, thus, magnitude of lattice energy will decrease from MgSO₄ to BaSO₄.
4(b)(i)	
4(b)(ii)	Enthalpy change of hydration (ΔH_{hyd}) of an ion is the energy evolved when <u>one mole</u> of the gaseous ion is hydrated under standard conditions. Ionic radius of the cation increases from Mg²⁺ to Ba²⁺. Since $ \Delta H_{hyd}(M^{2+}) \propto \left \frac{q^+}{r^+} \right $ Hence, ΔH_{hyd} decreases, i.e. ΔH_{hyd} becomes less negative.
4(c)(i)	Nucleophilic addition
4(c)(ii)	Hydrolysis
4(d)	
4(e)	Product from Step I:  Product from Step II: 

5(a)(i)	Denaturation refers to the <u>distortion of shape</u> of proteins by breaking the <u>weak bonds</u> holding the <u>secondary, tertiary and quaternary</u> structure but not the <u>covalent bonds</u> within the <u>primary</u> structure.
5(a)(ii)	<u>Heating</u> above 60°C will <u>disrupt</u> the <u>weak Van der Waals' forces</u> (and to a lesser extent, hydrogen bonds) holding the <u>quaternary, tertiary and secondary structures</u> , resulting in a <u>more disordered arrangement</u> . Or <u>Addition of acid / base</u> disrupt the <u>ionic bonds</u> holding the <u>tertiary and quaternary structures</u> .
5(b)	6 (do remember to check for duplicates)
5(c)(i)	$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \\ \text{H}-\text{N}-\text{C}-\text{C}-\text{O}-\text{H} \\ & & \\ \text{H} & & \end{array}$ and $\begin{array}{c} \text{O} \\ \\ \text{C}-\text{O}-\text{H} \\ \\ \text{H}-\text{N} \end{array}$
5(c)(ii)	
5(d)(i)	Acidic hydrolysis using <u>HCl (aq), heat over several hours</u> to break it down to its individual amino acids. (Or basic hydrolysis)
5(d)(ii)	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_4-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ $\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{NH}_2$
5(d)(iii)	Monomers from hydrolysis of protein has both <u>-NH₂ (amine group)</u> and <u>-COOH (carboxylic group)</u> whereas, monomers of Kevlar contained <u>only 1 of the functional group</u> in each of its monomer.
5(e)	$\text{HO}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\overset{\text{H}}{\underset{\text{COOH}}{\text{C}}}-\text{CH}_3$ $\text{CH}_3-\overset{\text{H}}{\underset{\text{OH}}{\text{C}}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
5(f)(i)	Acidic hydrolysis
5(f)(ii)	The products are able to form <u>favourable hydrogen bonds with water</u> , hence, is soluble in water.

Paper 3 Worked Solutions

1(a)(i)	Entropy of a chemical system measures the degree of disorder in a system. It gives a measure of the extent to which particles and energy are distributed within the system.
1(a)(ii)	- The mixing of Cl_2 and N_2 will result in an increase in the entropy of the system as there are now more ways in which the energy can be distributed among the molecules. - The heating of Cl_2 to a higher temperature means that there are more ways for the energy to be distributed among the molecules. Thus, the particles will move more randomly and at higher speeds. Thus, entropy will increase. - The chemical reaction between Cl_2 and I_2 to form ICl_3 will result in a decrease in entropy as there is a reduction in the number of molecules of gas in a chemical reaction by 1 mol. There are now less particles moving randomly and less ways to distribute the energy. - There will be an increase in entropy during photolysis as the number of particles in gaseous state increase by 1 mol. Each molecule is split into two chlorine radicals.
1(b)(i)	For reaction of hydrogen with chlorine, the reaction is explosive in sunlight, but slow in the dark below 200°C. For reaction of hydrogen with bromine, the reaction is slow at 300°C with Pt catalyst. The rate of reaction for H_2 with Cl_2 is faster than that between H_2 and Br_2.
1(b)(ii)	It is expected that the rate of reaction of hydrogen with fluorine will be higher . The reactivity of the halogen with hydrogen decreases down the Group as the oxidizing power of the halogens decreases down the group.
1(c)(i)	$\text{Cl}_2(\text{g}) / \text{Cl}_2$ in liquid CCl_4, uv light
1(c)(ii)	Free radical substitution Initiation step: $\text{Cl}_2 \rightarrow 2 \text{Cl}\cdot$ (slow step, uv light) Propagation step: $\text{Cl}\cdot + \text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$ $\text{C}_2\text{H}_5\cdot + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{Cl}\cdot$ Termination step: $2 \text{Cl}\cdot \rightarrow \text{Cl}_2$ $2 \text{C}_2\text{H}_5\cdot \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ $\text{C}_2\text{H}_5\cdot + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\text{Cl}$
1(c)(iii)	This is because in the first propagation step, in the abstraction of hydrogen radical from alkane to form HI, this reaction is highly endothermic . (Bond energy of HI = 299 kJ mol^{-1} vs Bond energy of HCl = 436 kJ mol^{-1}) Therefore, it is not possible for the reaction to occur.
1(d)	
1(e)(i)	CFCs are inert, non-toxic OR can be liquefied under pressure and is easily volatilised when pressure is released.
1(e)(ii)	CFCs, when reaching the ozone layer, undergoes homolytic fission between C-Cl bond under the action of uv radiation via photolysis. This forms Cl radicals which attack the ozone via a radical mechanism, resulting in a chain reaction of destruction of ozone molecules.
1(e)(iii)	Presence of C-H bonds in alkanes makes it flammable . Thus, alkanes may undergo combustion at high pressure conditions.

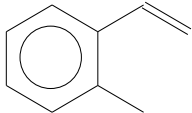
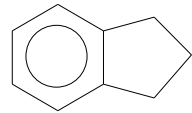
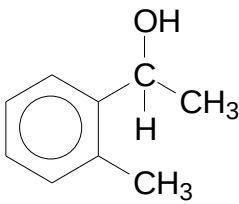
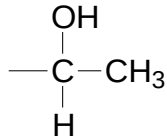
2(a)(i)	for a weak acid, $HA \rightleftharpoons H^+ + A^-$ where $K_a = \frac{[H^+][A^-]}{[HA]}$ $pK_a = -\log K_a$
2(a)(ii)	pH1: pH7: pH3: pH11: (Note: the electron-donating alkyl group on $-CO_2H$ reduces the acidic strength of $-CO_2H$, thus it will be protonated next)
2(b)(i)	<u>Quarternary</u> and <u>tertiary</u> structure of a protein will be altered. This occur because <u>denaturation</u> breaks the <u>weak bonds</u> holding the tertiary and quarternary structure, but not the covalent bonds within primary structure, resulting in a more disordered arrangement. Such action is normally done by salts, acids or enzymes. The weak bonds that are broken are the <u>side chain interactions</u>, which include <u>van der waals' forces of attraction</u>, <u>hydrogen bonds</u>, <u>ionic bonds</u> and <u>disulfide bonds</u>.
2(b)(ii)	It <u>disrupts</u> the <u>ionic bonds</u> between the charged R groups of the amino acids by forming new ionic bonds with the <u>carboxylate anions</u>.
2(b)(iii)	It <u>disrupts</u> the <u>ionic bonds</u> holding the tertiary and quarternary structures. The weak acid will dissociate to cause <u>carboxylate anions</u> or <u>amino groups</u> to be <u>protonated</u>. $-CO_2^- + H^+ \rightarrow -CO_2H$ $-NH_2 + H^+ \rightarrow -NH_3^+$
2(b)(iv)	$K_c = \frac{[gluconic\ acid]}{[GDL][H_2O]}$ Amount of GDL = $\frac{1.00}{178} = 0.005618 \text{ mol}$ Initial $[GDL] = \frac{0.005618}{\frac{50}{1000}} = 0.1124 \text{ mol dm}^{-3}$ $K_c = \frac{0.0670}{(0.1124 - 0.0670)(55.5)} = 0.0266 \text{ mol}^{-1} \text{ dm}^3$
2(c)(i)	OR
2(c)(ii)	Compound F Compound G

	Diadzein undergoes <u>reduction</u> of alkene and ketone with H₂ and Ni to form F. A <u>secondary alcohol</u> is formed. F has <u>three</u> hydroxyl groups, and thus able to react with three moles of sodium via <u>redox</u> reaction. F dissolves in NaOH(aq) as it contains phenolic groups which can undergoes <u>neutralization</u> with NaOH(aq) F undergoes <u>oxidation</u> with K₂Cr₂O₇ to form a <u>ketone</u> in G. G undergoes <u>condensation</u> with 2,4-DNPH to form hydrozone.
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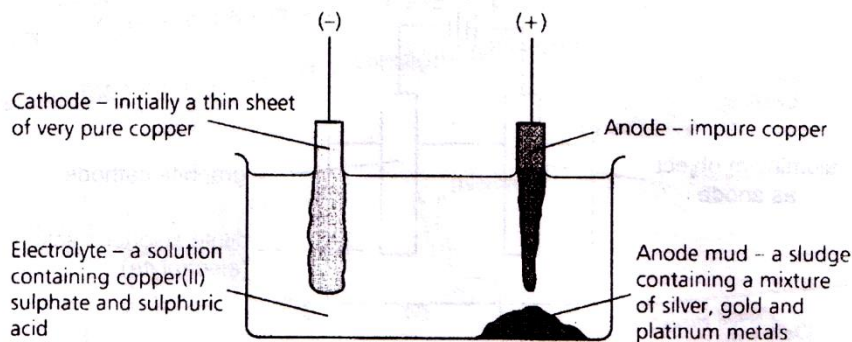
3(a)(i)	Cathode: O ₂ (g) + 4H ⁺ (aq) + 4e ⁻ → 2 H ₂ O (l)
3(a)(ii)	Cathode x 3 + Anode x 2 Overall: 2 CH ₃ OH + 3O ₂ → 2 CO ₂ + 4H ₂ O
3(a)(iii)	$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{red}} - E^{\circ}_{\text{CO}_2/\text{CH}_3\text{OH}}$ $1.18 = 1.23 - E^{\circ}_{\text{CO}_2/\text{CH}_3\text{OH}}$ $E^{\circ}_{\text{CO}_2/\text{CH}_3\text{OH}} = +0.05 \text{ V}$
3(a)(iv)	Methanol cell is <u>portable</u> , <u>easy storage</u> and <u>less flammable</u> as methanol is liquid. Thus, it need not be stored under pressure, unlike hydrogen gas.
3(b)(i)	CH ₃ CH ₂ OH(l) + 3O ₂ (g) → 2CO ₂ (g) + 3H ₂ O (l)
3(b)(ii)	$\Delta G = \Delta H - T\Delta S = (-1367000) - 298(-140)$ $= -1339280 \text{ J mol}^{-1}$ $= -1340 \text{ kJ mol}^{-1}$
3(b)(iii)	Oxidation: CH ₃ CH ₂ OH(l) + 3H ₂ O(l) → 2CO ₂ (g) + 12H ⁺ (aq) + 12e ⁻ $E^{\circ}_{\text{cell}} = -\frac{\Delta G^{\circ}}{zF}$ $z = 12$ $E^{\circ}_{\text{cell}} = -\frac{-1339280}{12(96500)} = +1.16\text{V}$
3(c)(i)	$\Delta H = \text{bond breaking} - \text{bond forming} = [(C-C)+5(C-H)+(C-O)+(O-H)+3(O=O)] - [4(C=O)+6(O-H)]$ $= [(350)+5(410)+(360)+(460)+3(496)] - [4(805)+6(460)]$ $= -1272 \text{ kJ mol}^{-1}$
3(c)(ii)	The enthalpy change of vapourisation of water and ethanol is not considered in the calculation for (c)(i) OR The <u>bond energies</u> are of <u>average values</u> .
3(d)(i)	$\text{C}_x\text{H}_y\text{OH} + \text{Na} \rightarrow \text{C}_x\text{H}_y\text{O}^- \text{Na}^+ + \frac{1}{2} \text{H}_2$ Amount of H ₂ produced = $\frac{1}{2}$ (Amount of J used) $= \frac{10.9}{24000} = 0.0004542 \text{ mol}$ Amount of J used = 0.0009084 mol $\text{C}_x\text{H}_y\text{OH} + \left(x + \frac{y-1}{4}\right) \text{O}_2 \rightarrow x\text{CO}_2 + \left(\frac{y+1}{2}\right) \text{H}_2\text{O}$ Amount of CO ₂ absorbed by NaOH(aq) = $\frac{109}{24000} = 0.004542 \text{ mol}$ Comparing mole ratio C _x H _y OH ≡ xCO ₂ $0.0009084 : 0.004542$ $1 : 5$ $x = 5$ In the combustion of the alcohol, V(O ₂) used - V(CO ₂) produced = 54.4 cm ³ (i.e. reduction in volume) V(O ₂) used = 109 + 54.4 = 163.4 cm ³ Amount of oxygen used = $\frac{163.4}{24000} = 0.006808 \text{ mol}$ Comparing mole ratio $\left(x + \frac{y-1}{4}\right) \text{O}_2 \equiv x\text{CO}_2$

	$\left(1 + \frac{y-1}{20}\right) \text{O}_2 \equiv \text{CO}_2$ $0.006808 : 0.004542$ $1.50 : 1$ $1.5 = \left(1 + \frac{y-1}{20}\right)$ $y = 11$
3(d)(ii)	J undergoes <u>oxidation</u> with $\text{K}_2\text{Cr}_2\text{O}_7$. J must be either <u>primary or secondary alcohol</u>. J undergoes <u>elimination</u> to form alkene K K undergoes <u>vigorous oxidation</u> to result in the cleavage of the alkene bond to form ethanoic acid and propanone. $\begin{array}{c} \text{CH}_3 \quad \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \quad \text{H} \end{array}$ K $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ J
3(d)(iii)	Concentrated H_2SO_4 , reflux at 170°C
3(d)(iv)	K will not show geometrical isomerism as there are <u>two methyl groups</u> attached on the <u>same carbon</u> involved in the <u>C=C bond</u>. <i>(Note: Statements that are not acceptable:</i> (1) <i>Methyl groups are on the same side of the double bond.</i> (2) <i>There is a carbon with two methyl groups)</i>

4(a)	Nucleon number is the <u>total number of protons and neutrons</u> in a <u>nucleus of an atom</u> .
4(b)	In an electric field, the <u>electron</u> will be <u>deflected</u> to the <u>positive plate</u>, while the <u>proton</u> will be <u>deflected</u> to the <u>negative plate</u>. In an electric field, the <u>angle of deflection</u> for an electron is <u>larger</u> than that of the proton as it is lighter than the proton.
4(c)(i)	$\begin{array}{c} \times \times \quad \cdot \quad \times \times \\ \times \text{O} : \text{N} : \text{O} \times \\ \times \times \quad \times \quad \times \end{array}$ NO_2 $\begin{array}{c} \cdot \cdot \quad \times \times \quad \cdot \cdot \\ : \text{O} \times \text{O} \times \text{O} : \\ \cdot \cdot \quad \times \quad \times \end{array}$ O_3
4(c)(ii)	For O_3, there are 2 bond pairs and 1 lone pair. The 3 electron pairs will point towards the corners of an equilateral triangle. The <u>lone pair-bond pair repulsion</u> > <u>bond pair-bond pair repulsion</u> because the <u>lone pair</u> is closer to the oxygen than a bond pair. Thus, bond angle is 115° with the shape being bent. For NO_2, there are 2 bond pairs and 1 unpaired electron. The 3 electron pairs will point towards the corners of an equilateral triangle. The <u>unpaired electron-bond pair repulsion</u> is <u>weaker than that of the lone pair-bond pair repulsion</u> in O_3. Thus, bond angle is <u>(any value between O_3 and 170°)</u> with the shape being bent.
4(c)(iii)	F, being a period 2 element, is <u>not able to expand</u> to accommodate more than 8 valence electrons. To form FO_2, F will <u>need to form two dative bonds</u>. Additionally, F is <u>too electronegative</u> form dative bonding. Cl, being a period 3 element, is able to use its <u>3d orbitals</u> to <u>accommodate more than 8 valence electrons</u> through the use of energetically accessible 3d orbitals.
4(d)(i)	A <u>sample of calcium</u> is being held on a <u>platinum wire</u> that has been repeatedly cleaned with hydrochloric acid to remove traces of previous analytes. The sample is then <u>burnt</u> in a hot, non-luminous flame in the <u>presence of oxygen</u> and the colour is being observed. A <u>brick red flame colour</u> is observed.
4(d)(ii)	Oxides of Group II elements will react with water to form hydroxides. As the Group II metal hydroxides goes down the group from magnesium to barium, the <u>solubility of the hydroxides increases</u>. This increases the <u>extent of dissociation of the hydroxides</u>, thus resulting in the <u>increase in the concentration of hydroxide ions</u>, and therefore the increase of the pH value from Mg to Ba.

4(e)	$\text{Ba}(\text{O}_3)_2 + \text{H}_2\text{O} \rightarrow \frac{5}{2} \text{O}_2 + \text{Ba}(\text{OH})_2$
4(f)(i)	Amount of $\text{Na}_2\text{S}_2\text{O}_3 = \frac{15}{1000} \times 0.100 = 0.001500 \text{ mol}$ Amount of $\text{I}_2 = 0.000750 \text{ mol}$
4(f)(ii)	Amount of $\text{O}_3 = 0.000750 \text{ mol}$ Volume of $\text{O}_3 = 0.000750 \times 22.4 = 0.0168 \text{ dm}^3$ % $\text{O}_3 = \frac{16.8}{500} \times 100\% = 3.36\%$
4(g)(i)	For production of aldehyde, use <u>acidified $\text{K}_2\text{Cr}_2\text{O}_7$ and heat/reflux with distillation.</u> For production of carboxylic acid, use <u>acidified $\text{K}_2\text{Cr}_2\text{O}_7$ and heat/reflux</u>
4(g)(ii)	L:  From molecular formula, compound is unsaturated. To be able to react with $\text{Br}_2(\text{aq})$ indicates that a double bond is present, capable of undergoing electrophilic addition. M:  Molecular formula is the same as L, indicating that it must contain a cycloalkane in the compound as it does not react with $\text{Br}_2(\text{aq})$, indicating the absence of an alkene bond. N:   N must undergo oxidation with alkaline aqueous iodine, thus having the structure

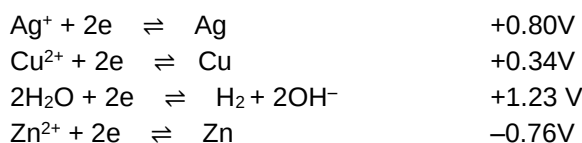
5(a)



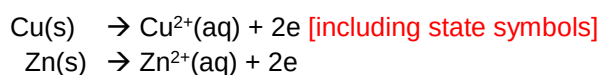
In electrochemical purification of copper

Anode: Crude copper; **Cathode:** Pure copper; **Electrolyte:** Copper (II) sulphate

From the Data Booklet:

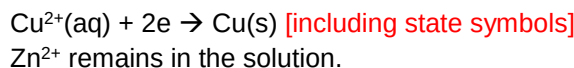


When an electric current is applied, both zinc and copper are oxidised to Zn^{2+} and Cu^{2+} respectively since E° of Zn^{2+}/Zn is less positive than the E° of Cu^{2+}/Cu .



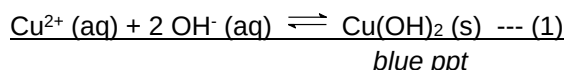
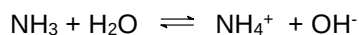
Silver, due to its more positive (E°) reduction potential will not be oxidised.
Silver will drop to the electrolytic bed as elemental silver.

Zn^{2+} and Cu^{2+} migrate to the cathode.
Only Cu^{2+} is reduced to copper due to its more favourable reduction potential and higher concentration.



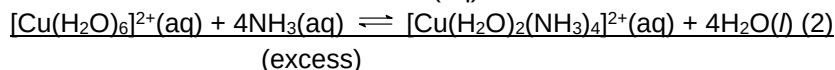
5(b)

When NH_3 is first added to $\text{Cu}^{2+}(\text{aq})$:



Ionic product, $[\text{Cu}^{2+}][\text{OH}^-]^2 > K_{\text{sp}}$ of Cu(OH)_2
Blue ppt of Cu(OH)_2 is formed.

When excess NH_3 is added to $\text{Cu}^{2+}(\text{aq})$:



NH_3 ligands replaces the H_2O ligands, forming a more stable deep blue $[\text{Cu(NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ complex with $\text{Cu}^{2+}(\text{aq})$.

$[\text{Cu}^{2+}]$ decreases as it is being used to form the complex
equilibrium position in (1) shifts left to increase $[\text{Cu}^{2+}]$

Ionic product, $[\text{Cu}^{2+}][\text{OH}^-]^2 < K_{\text{sp}}$ of Cu(OH)_2
Pale blue ppt, Cu(OH)_2 , dissolves.

5(c)	$\text{O}^{2-} + \text{NH}_3 \rightarrow \text{NH}_2^- + \text{OH}^-$ (note: acid is proton donor while base is proton acceptor) The colour of the solution is <u>dark blue</u> in colour.
5(d)	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{CuCl}_4]^{2-} (\text{aq}) + 6 \text{H}_2\text{O} (\text{l})$ <u>Ligand exchange</u> reaction has occurred. $[\text{Cl}^-]$ increases in concentrated hydrochloric acid, <u>equilibrium position</u> will shift <u>right</u> to form <u>yellow</u> $[\text{CuCl}_4]^{2-}$ complex. Presences of both $\text{Cu}^{2+} (\text{aq})$ and $[\text{CuCl}_4]^{2-}$ in the equilibrium mixture results in a <u>yellow-green</u> solution. Solution remains blue when concentrated sulfuric acid is added as the sulfate ions are weaker ligands. Dilution of the yellow-green solution will <u>reduce the $[\text{Cl}^-]$</u>, hence by LCP, the <u>position of the equilibrium</u> will shift to the <u>left</u>, producing the original pale blue colour.
5(e)(i)	$4\text{I}^- + 2\text{Cu}^{2+} \rightarrow \text{I}_2 + 2\text{CuI}$
5(e)(ii)	$E_{\text{cell}} = -0.39 \text{ V}$ Reaction is thus <u>not feasible</u> .
5(e)(iii)	CuI is <u>insoluble</u> in aqueous solutions. $\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$ Thus $[\text{Cu}^+]$ will be <u>very low</u>. Therefore, by LCP, the position of equilibrium will shift <u>towards the production of CuI</u>, hence resulting in the favouring of the forward reaction.
5(f)(i)	<u>Pale blue solution will decolourised</u> , and a <u>brick-red precipitate</u> will be produced.
5(f)(ii)	Positive result: $\begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \text{O} \\ & & & & & & // \\ & & & & & & \text{H} \end{array}$ $\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{O} \\ & & & & & & & // \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{C} & - & \text{H} \\ & & & & & & & \\ & & & & & \text{H} & & \\ & & & & & & & \text{H} \end{array}$ Negative result: $\begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{O} & \text{H} & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & & \text{H} & \end{array}$

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